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CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "**Intelligent Network Intrusion Detection System**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Dr. Prabhu K.** School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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ABSTRACT

Due to the fast development of interconnected networks and IT-solutions, the level of complexity of the cyberattacks has significantly increased, and it is necessary to find innovative solutions for detecting attacks. This project suggests the development of an Intelligent Network Intrusion Detection System (INIDS) based on application of machine learning methods in order to classify and detect attacks on the network. It includes monitoring of **network traffic, extraction of network features** like connection time, protocol used, number of packets etc., in order to differentiate normal and abnormal activities of the network. Machine learning models are trained on the basis of the dataset representing network traffic, and attacks of different kinds like **Denial of Service, Probing and Access are detected.** The suggested IIDS allows avoiding false detections and providing security administrators with real-time alerts.

The proposed methodology involves a structured framework involving **data acquisition, data preprocessing, feature extraction, model training, and attack classification.** Network data obtained is cleaned and formatted appropriately in order to feed into machine learning algorithms in an effort to detect patterns related to malicious activity. This system is designed to continuously monitor network traffic and create an alert whenever suspicious or harmful traffic is found. In addition to the improvement in terms of adaptability, this proposed solution is intended to help the intrusion detection system learn from existing patterns of attacks and differentiate these attacks from normal network activity. System performance is measured by traditional performance metrics including **accuracy, precision, recall, and F1-score.**

1. Problem Definition

Due to the fast-growing computer networks, cloud computing, and Internet of Things (IoT), the possibilities for cyberattacks and access are increasing. However, attackers are constantly coming up with new methods of conducting cyberattacks that can take advantage of certain flaws in software products, rendering traditional security measures ineffective in detecting such malicious actions. **Conventional intrusion detection** systems use a set of rules and signatures that can only detect attacks that have already been discovered. The problem is that they might be generating too many **false alarms**.

As such, the need exists for an **Intelligent Intrusion Detection System (IIDS)** that can make use of machine learning methods to study network traffic and detect any abnormal

behaviour. The IIDS must have the ability to process relevant network characteristics, learn from past traffic records, and classify the network activities into either normal or malicious. Moreover, it will have to identify the type of attack such as DoS, probing, and unauthorized access among others and raise alerts when any form of suspicious activity is observed.

The major issue being addressed by this project is that of developing an efficient and intelligent intrusion detection system which not only helps in the detection of attacks in networks but also **minimizes false positives.**

2. Literature Survey

2.1 From Traditional IDS to Intelligent IDS

An Intrusion Detection System (IDS) acts like a security guard for a network: it watches traffic. Tries to decide whether an activity is normal or dangerous. Traditional IDS mainly follow two paths. Signature-based systems look for known attack patterns so they are effective against threats but weak against new or rare attacks. Anomaly-based systems learn what normal behaviour looks like, but changing network behaviour can make them produce many false alarms. This limitation has turned research toward Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL). The literature shows that the goal has gradually changed.

Earlier work mainly asked, "Can the system detect an attack?" Modern research asks a harder question: "Can it detect the attack accurately explain why it happened, protect private data adapt to new attacks and still run fast enough in the real world?"

2.2 Machine Learning and Deep Learning: Teaching the IDS to Recognise Patterns

ML methods such as Decision Trees, Random Forests, Support Vector Machines and Naïve Bayes are relatively simple and useful for intrusion classification. However network traffic is not simple. It contains numbers of features and complicated relationships. Deep learning models such as **CNN, RNN, LSTM, GRU, MLP, DNN and Autoencoder** can learn these patterns automatically reducing the dependence on designed rules.

The reviewed studies show that there is no best" model for every environment. A CNN can offer a balance between accuracy and computational cost while models such as

CNN-BiLSTM may capture more complex patterns but require more resources. Olanrewaju-George and Pranggono also found results from an unsupervised Autoencoder for IoT intrusion detection. Other work improves the learning process by selecting useful features or optimizing model parameters. In one reviewed study feature selection combined with learning achieved **95.59 % detection accuracy** while another federated deep-learning approach reported 98.24 % accuracy and **86.77 % F1-score.**

2.3 Transformers: Looking at the Bigger Picture

CNNs are good at finding patterns and recurrent models are useful for sequences but network traffic can contain relationships that stretch across many features or events. Transformers approach this problem through self-attention: of treating every part of the traffic as equally important the model can focus on relationships that matter most.

The literature reports Transformer-based IDS approaches. **IDS-MTran** combines traffic features with Transformer processing and achieved 99.25 % binary-classification accuracy on **NSL-KDD** and 99.16 % on its five-class task. **MBConv-ViT** combines convolution with a Vision Transformer and reported 97.14 % accuracy on **ToN-IoT** and 99.99 % on Bot-IoT. However these numbers should not be treated as proof of superiority: Transformer models can be computationally expensive and minority attacks may still be detected poorly when their training data is limited.

2.4 The Privacy Problem: Why Centralised Learning Starts to Break

A intelligent IDS usually sends large amounts of network data to a central location for training. This is convenient. It creates a serious trade-off: the more data the system collects the more sensitive information it may hold in one place. Privacy, communication overhead and the risk of a failure therefore become important concerns, especially in IoT networks.

Federated Learning (FL) changes the old way. Each device trains a model locally. Sends model updates rather than raw traffic. A server combines those updates into a shared model. Sends the improved model back. This allows collaboration without pooling all raw data.

2.5 Federated Learning: Smarter Collaboration, New Problems

The reviewed studies show that FL can improve privacy while maintaining detection performance. In **PEIoT-DS FedAvgM** achieved 95.05 % accuracy compared with 94.05 % for FedAvg. It also reached 90 % accuracy in communication rounds and reduced the false-positive rate from 5.1 % to 3.9 %. These results show why aggregation strategy matters than treating federated learning as a single fixed technique.

Fl is not a magic privacy button. Its biggest practical problem is data heterogeneity. Different devices see traffic so their local datasets are often non-IID. The **Edge-FL** study makes this problem very clear: its lightweight 1D-CNN achieved around 99.9 % accuracy in cases yet accuracy fell to 52.39 % for a difficult 12-class non-IID setting with seven clients. In words a model can look almost perfect in a controlled experiment and still stumble when the network becomes messy and uneven.

2.6 Lightweight IDS: Putting Intelligence Where the Traffic is

IoT devices do not have CPU power, memory, battery or bandwidth. A huge model may be highly accurate on a computer but impractical on a sensor, vehicle, gateway or industrial device. This is why Edge Computing is increasingly connected with FL. Of sending everything to a distant cloud processing can move closer to where the network traffic is produced.

The literature therefore emphasizes architectures, especially compact CNN-based models together with edge processing and federated training. This combination aims to reduce latency and communication while keeping the data local. The trade-off must be made. Inversely reducing size can significantly effect learning capability and increase complexity can increase processing power, energy consumption and delay.

2.7 Imbalanced Traffic when Accuracy can be false

Another common problem is class imbalance. A dataset can be dominated by normal traffic and common attacks, while rare attacks have few examples. Thus, a model can report overall accuracy while silently failing at the attacks that matter most. In the second survey, SMOTE is used to generate samples for the minority class. This can improve attack representation, but synthetic data also add processing requirements and must be handled carefully in privacy-sensitive environments.

2.8 Explainability and Robustness

A security system cannot depend on high accuracy alone. If an IDS reports “attack detected” but does not explain why an analyst has reason to trust the alert. Explainable AI attempts to open this box. One reviewed approach combines transformer-based learning, **SMOTE and CNN-LSTM** detection with AI while **HADA-FL** uses **SHAP** to identify the features that contribute most to a decision.

Explainability also connects naturally with robustness. Modern ML-based IDSs can themselves be attacked. Small changes in input traffic may cause a model to classify activity as normal. HADA-FL therefore combines SHAP-guided feature selection with privacy, secure aggregation and testing against attacks such as **FGSM, PGD and label-flipping.** Its results show that robustness, privacy and explainability are becoming parts of the problem rather than separate research topics.

2.9 Adaptability: The Network never stands still

A network is not an environment. Devices. Disappear traffic patterns change and attackers continuously modify their techniques. A model trained on old traffic can gradually become outdated. The second survey therefore introduces learning, where a model learns new information while attempting to preserve previously learned knowledge.

This becomes especially important when combined with learning. The aim is to create a system that can learn from attacks without completely forgetting old ones a problem known as catastrophic forgetting. The literature suggests that adaptive and federated learning together could make IDS more suitable for changing, distributed networks although efficient implementation in environments remains unresolved.

2.10 Connected Research Gaps

Putting results of both surveys side by side the gaps create a picture and not a number of unrelated issues:

Accuracy versus reality: really high benchmark accuracy is not directly synonymous to high real-word performance.

Centralization versus privacy: It is easy to train all data in one place. But not so practical to share the raw traffic information which can contain sensitive data.

FL versus Heterogeneity: FL guarantees data locality. Clients Non IID traffic could really make the convergence and performance collapse.

Complexity versus deployment: These sophisticated Transformers or Hybrid architectures could require a lot of GPU memory, energy and communication capability.

Accuracy versus Attacks: Minor minority threats can be unnoticed in systems that show amazing overall accuracy since the class-distribution is class-imbalanced.

Detection versus trust: Explainability has to be there so an analyst can trust the reasons behind the IDS alarm.

Learning versus change: Training needs to be incremental as attack pattern and network behavior changes constantly.

2.11 Overall Direction for an IDS

The results of both studies suggest a similar architecture for intelligent IDS of the future. As learning or Transformer-based feature learning the model for correct detection, federated learning so that traffic will be kept private, and edge or lightweight processing so the latency is minimal and resources can be preserved are essential. The attacks can then be offset by SMOTE or another balancing algorithm, SHAP or e-explainable AI will further make the model's decisions understandable, and finally incremental and robust learning is required so the system does not fall prey to evolving networks and attacks.

The key observation remains: no longer is an IDS represented solely by its model with a correct accuracy.

It's a system that keeps accurate despite uneven data and shifting adversaries, explains its decisions to an analyst, secures information while learning, and still satisfies network constraints. That's what constitutes a valuable IDS system, not just the model.

2.11 Conclusion of the Survey

The literature review found that intelligent intrusion detection is shifting from attack detection toward learning-based systems that can protect data and adapt to security threats and analysts. Deep learning and Transformers allow system to better detect network traffic. Federated Learning supports protecting network data. Edge computing and lightweight model bring practicality to resource-constrained systems. Explainable and incremental learning can help the systems.The main challenge is to combine these strengths without making IDS too complex or slow. A practical intelligent IDS should balance **accuracy, privacy, efficiency, explainability, robustness and adaptability** of focusing on accuracy alone.

3. Objectives

The main objectives of the Intrusion detection system are-

1.To develop an efficient Intrusion detection system which can accurately identify **malicious network activities**.

2.To have a better detection of diverse cyberattacks like **DoS or DDoS, scanning or brute force.**

3.To address **imbalance of class** in intrusion datasets and improve detection of less

happening attack classes.

4.To make it computationally **lightweight and efficient** for detection so that processing is reduced.

5.To make detection better for evolving and previously **unseen attacks**, by methods like anomaly detection.

6.To evaluate IDS using multiple datasets and realistic **network traffic** .

7.To reduce the chances of **false positives and false negatives** to increase reliability.

8.To provide **security alerts** that helps users to understand why network activity is suspicious.

9.To evaluate the system by using standard performance metrics, like **accuracy , precision , recall, f1 score and computational efficiency**.

4. Scope

The **INIDS** project scope is the development of a machine learning system for monitoring the network traffic and identifying the activities that may represent a cyberattack. The project mainly involves the analysis of the network traffic data to extract its features and classify it as benign or malicious.

The main areas addressed by the project are:

Network Traffic Monitoring: The system will examine the network traffic data to detect any anomalies in the network.

Data Preprocessing: The network traffic data will be preprocessed, where irrelevant data is removed and the features are converted into the appropriate form.

Feature Selection: The features of the network, including protocol type, duration, packets, and connectivity behavior, will be selected in order to detect any anomaly in the network.

Detection and Classification of Attacks: The system will classify network traffic and attacks on the network, including Denial-of-Service (DoS) attack, probe attack, and access attempt, among others.

Machine Learning Model: Machine learning algorithms will be trained and tested to predict the feasibility in detecting malicious network traffic accurately. Performance Evaluation: The developed models will be tested by performance metrics such as accuracy, precision, recall, and F1-score to evaluate the performance for detection.

Alert Generation: The system will provide an alert when suspicious or potentially malicious network activity is detected, helping the user or security administrator take further action.

Future Scalability: The system can later be extended to support real-time network traffic monitoring, larger datasets, additional attack categories, and more advanced machine learning or deep learning techniques.

This project is mainly geared towards network level intrusion detection rather than being designed to be used as a replacement for firewalls, anti-virus software or even as SIEM solutions. It is basically meant to act as an intelligent layer of security that will help in detecting any suspicious activity on the network.

5. Feasibility Study

5.1 Technical Feasibility

The proposed IDS can be developed using commonly available programming languages, machine learning libraries, datasets, etc. It would be possible to process network traffic/intrusion datasets to extract relevant features which can then be classified into normal/malicious activity.

Various ML/DL techniques have been discussed which could be utilized for intrusion detection purpose. However, by utilizing suitable preprocessing/feature-selection techniques, we can alleviate the problem of increased computational complexity. As revealed from our research survey, some important directions for further improvements over existing IDS approaches include application of lightweight models, utilization of efficient feature selection techniques, as well as testing them on real world scenarios. Hence, the proposed system would try to achieve an appropriate balance between these factors.

5.2 Operational Feasibility

The proposed system intends to provide an automatic detection mechanism for detecting suspicious network activities along with generating easily understandable security alerts which would allow the users or Network Administrators to identify potential threats.

User-Friendly Interface Designing the proposed Intrusion Detection System (IDS) with user friendly interface that displays various types of attacks and related security informations helps in providing easier and faster way for Identification of Potential

Security Incidents.

5.3 Economic Feasibility

The proposed system will mainly consist of an application that could primarily develop with some open-source programming language, Machine Learning Frameworks, and intrusion detection dataset that are publically available.

There won't be a need for expensive specialized hardware at least initially for its Implementation. Also, using Lightweight and Computationally Efficient Models would further bring down the Infrastructure as well as Processing Requirements. Thus, we might expect the overall Development Cost of the Proposed System to be on the Lower Side despite having Some Usefull Intrusion Detection Capabilities.

5.4 Schedule Feasibility

This Project can easily be done as it is planned in many Stages which are Literature Study & Requirement Analysis, Dataset Collection and Preprocessing, Feature Selection, Model Development, System Implementation, Performance Evaluation & Documentation etc.

All these Components of this Modular based Proposed IDS can be Developed & Tested Individually then Integrates into Complete System.

Taking into account various aspects (technical, operational, economic, schedules), the suggested intrusion detection system is feasible.

The primary goal is to overcome the shortcomings found in the studies that have been conducted thus far: computational effectiveness, class unbalance, restricted number of attacks, lack of real-word assessment, and detecting evolvement threats. So the developed strategy may become a basis for creating effective and efficient IDS.

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